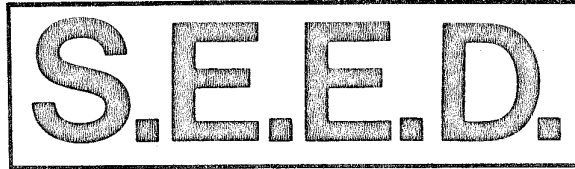


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Scans of items related to `Kansas City` type Cassete I/O:-

S.E.E.D. CI30/60 Cassete tape interface document
+ Read and write curcuit diagrams
+ Connection diagram

544



STRUMECH ENGINEERING ELECTRONICS DIVISION

CI 30/60

The CI 30/60 Cassette interface is a low cost, reliable method of storing and loading your programmes, taking advantage of your RS -232C terminal port.

COMPATIBLE RECORDING STANDARD

The recording method is compatible with the so-called "Kansas City" standard, first to keep it simple and secondly to promote standardization. Under this standard, when operating at 300 baud, a mark or logic '1' will be recorded as 8 cycles of 2400 Hz while a space or logic '0' will be recorded as 4 cycles of 1200 Hz. The data will be stored as frequency encoded ASCII Code.

You will notice that a UARI does not appear in the diagrams. This is because this simple system utilizes the UART or UART-like device already available on your computer's terminal I/O port. Assuming that the computer system in question has an RS-232C type interface, the arrangement shown will enable the cassette system to communicate with the computer. This system requires only that the UART transmit-and-receive clock signals be separable either by unsoldering or breaking a PC trace, and requires only the addition of the DPDT switch supplied.

SYSTEM CHECK OUT

The system can be tested by connecting the 'AUX' output to the input of the recovery section, eliminating the recorder from the loop. Connect a 4800 Hz clock to the clock line and then alternately connect the data in line to a logic '0' and a logic '1'. The data out should reflect the state of the data in, if not, something has gone astray. If you have access to an oscilloscope, the 'clock out' line can also be checked.

The use of the system is as simple as its construction. Assuming that you have a low speed paper tape load-and-punch program similar to that provided in the Motorola MIKBUG * ROM firmware, you can simulate paper tape operations. Before initiating the punch function, simply place the tape recorder in the record mode and place the cassette interface switch in the record position. As the data flows to the terminal it will also go to the record section of the cassette system.

To load data from tape initiate the load function from the keyboard, place the cassette interface switch in the playback position and switch the recorder to the playback mode. Data flows to the computer as if the source were the paper tape reader.

RELIABILITY

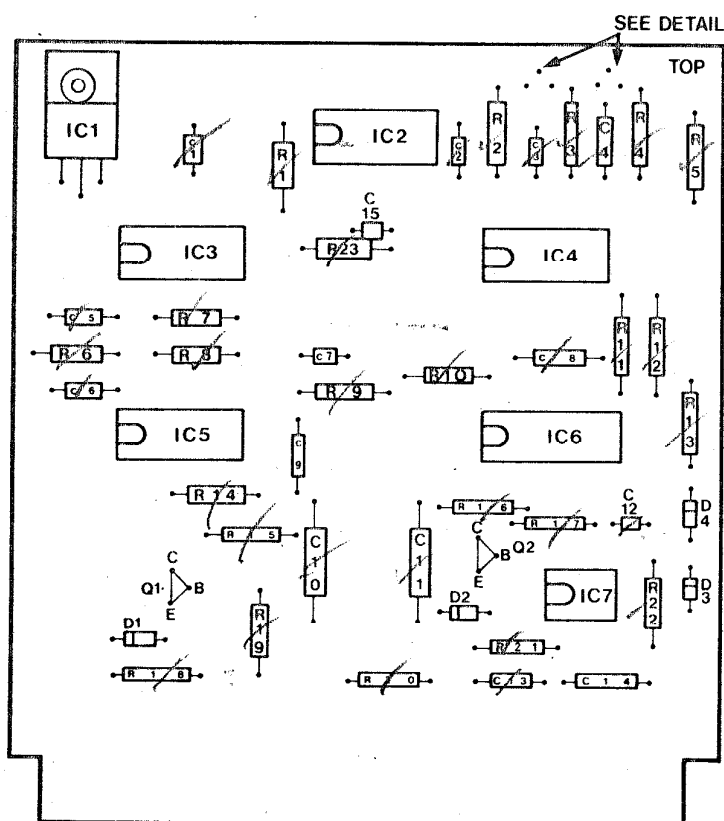
The CI-30/60 cassette interface is a relatively non-critical, wide tolerance system, it has been thoroughly tested and exhibited no detected errors.

To obtain the widest speed tolerance however the pulse width at pin 13 of IC6 should be adjusted with the aid of an oscilloscope to 277.3 Micro sec with logic '0' on the input.

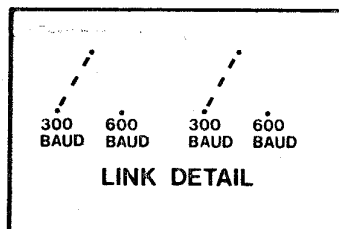
In order to keep error rates to a minimum the following precautions should be observed :-

- 1) Use a good quality cassette recorder.
- 2) Keep the transport and recording head immaculately clean.
- 3) Use fresh batteries or better still use the A.C. mains adaptor.
- 4) Always use a good quality tape.

For those interested in a faster transfer rate, this system has undergone some preliminary tests at 1200 baud and seems to perform as well as at 300 baud.



COMPONENT LAYOUT



LINK DETAIL

CI 30/60 COMPONENT LIST

✓R1	100	✓C1	0.1 μ f
✓R2	33K	✓C2	0.001 μ f.
✓R3	10K	✓C3	0.1 μ f.
✓R4	4.7 K	✓C4	0.047 μ f.
✓R5	1 K	✓C5	.1 μ f.
✓R6	100	✓C6	.1 μ f.
✓R7	100 K	✓C7	.01 μ f.
✓R8	10 K	✓C8	.001 μ f.
✓R9	1 K	✓C9	.1 μ f.
✓R10	1 K	✓C10	.47 μ f.
✓R11	33 K	✓C11	.22 μ f.
✓R12	33 K	✓C12	.001 μ f.
✓R13	1-2 K	✓C13	.1 μ f.
✓R14	4-7 K	✓C14	0.01 μ f.
✓R15	470	✓C15	.1 μ f.
✓R16	47K	Q1	2N2222
✓R17	47K	Q2	2N2222
✓R18	10 K		
✓R19	470	D1-D4	1N4148
✓R20	27K		
✓R21	30K	IC1	7805
✓R22	10K	✓IC2	SN74123
✓R23	1K	✓IC3	SN74107
		✓IC4	SN7400
		✓IC5	SN7474
		✓IC6	SN74123
		IC7	741C ?
		RLY	RELAY

EDGE CONNECTOR SOCKET

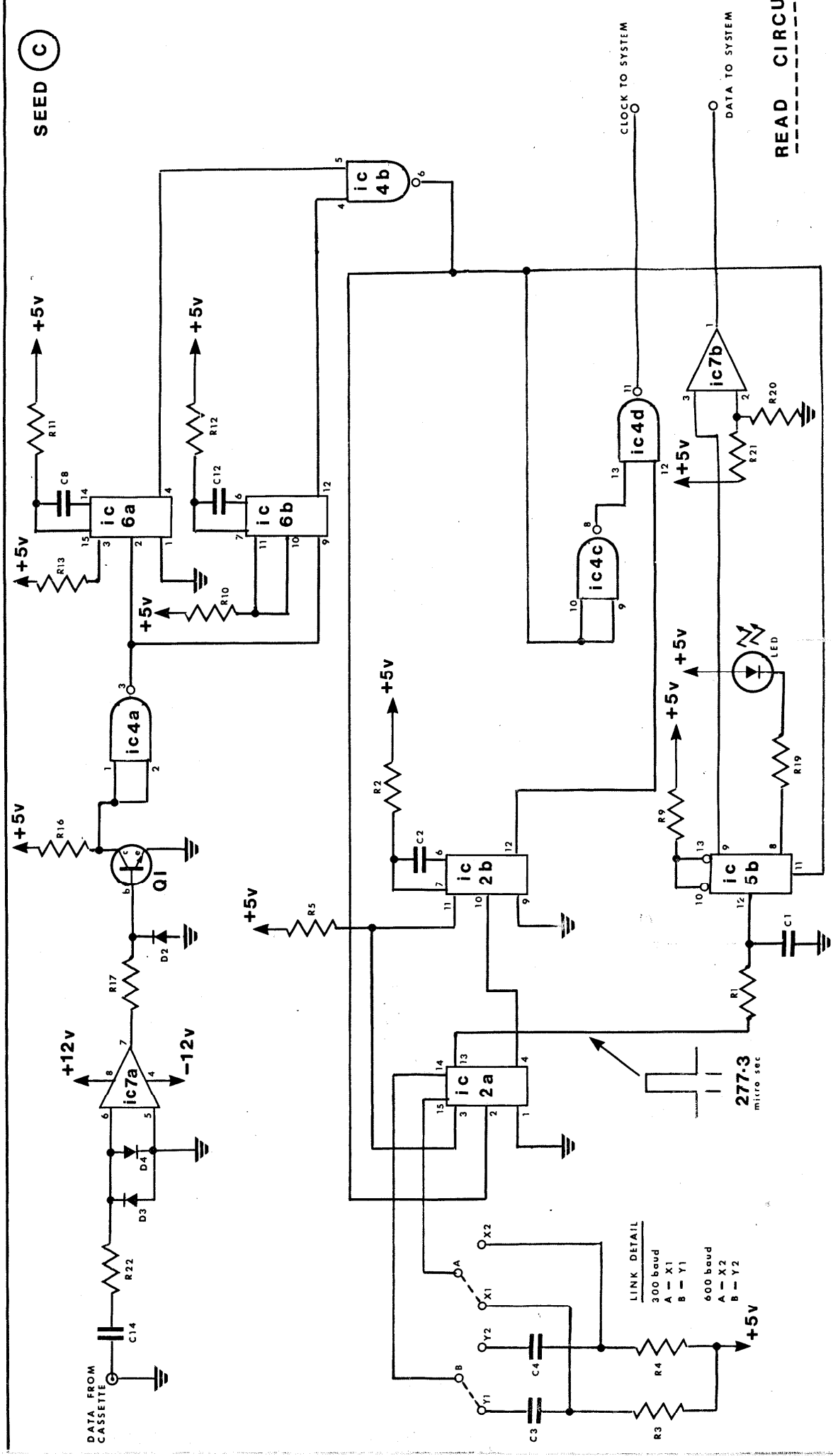
L.E.D.s.

1.5 Mts. 10 Core Ribbon cable

Assembly Instructions

- 1) All components must be inserted from the top of the board, and soldered on the bottom surface only.
- 2) Insert Diodes D1-D4 and solder. Care must be taken to ensure that they are correctly orientated. (See component layout).
- 3) Install all resistors and solder.
- 4) Install all capacitors and solder.
- 5) Install integrated circuit IC1, this should be inserted so that the flat surface is in contact with the circuit board as this acts as heat sink.
- 6) Install the remaining IC's taking care to insert them correctly.
- 7) Decide on which Baud rate is required and Jumper the board accordingly.
- 8) Insert pin into all remaining holes on board and solder top and bottom surfaces.
- 9) Now solder all remaining connections on top of board.
- 10) Insert P.C.B. in edge connector socket with components face down and wire up as shown.

SEED (C)



READ CIRCUIT

